

BSc (Hons) Computing / BSc (Hons) Computer Science
/ BSc (Hons) Artificial Intelligence

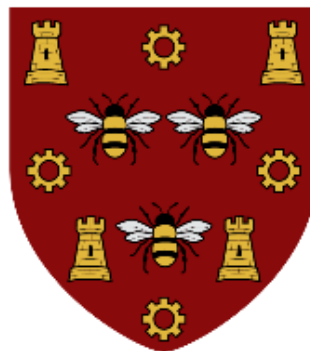
COM5409

Web Design and Programming

MODULE GUIDE FOR BLOCK DELIVERY

2025/2026

Semester 2



Level HE5

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1. Key Information

Module Tutor	Chiara Fasching
Module Tutor Email	c.fasching@greatermanchester.ac.uk
Other Contact Methods	COM5409-Web Design and Programming Teams Channel
Availability	I'm definitely available in person Tuesday 13:30 -14:00 & Friday 13:30 – 14:30 . For other times or online meetings, send an email.
Weblink to Moodle Class	https://moodle.greatermanchester.ac.uk/course/view.php?id=36335
Weblink to Module Specification	See https://modules.bolton.ac.uk/COM5409

2. Module Communications

The Module Tutor's contact details are provided above. You must check your **University of Greater Manchester email address** and the **Moodle** area dedicated to this module regularly as many module communications are channelled through these media.

A [Microsoft Teams Group](#) should be joined by using the join code **opx7ozq**. Each cohort has an assigned channel for streamline communication and will be used regularly to broadcast class activities. Announcements will be broadcast via Moodle and/or Teams.

Your Module Tutor will normally aim to respond to your email messages within **2 full working days** of receipt. However, responses will be longer in holiday periods.

3. Module Description

In this module, you will look at how to design and code as a web developer, thus developing skills in web design and implementation in terms of UI and UX considerations, styling, client-side coding, server-side coding with database access using contemporary methods, tools, languages, frameworks, and standards in addition to issues surrounding web technology. As requirements have increased, so too have user expectations of both the design and functionality of such solutions. During the early years of the world-wide-web, web designers focused mainly on creative concerns of how sites looked and functioned from a User Interface (UI) and User Experience (UX) perspective, while programmers produced code making the sites interactive and dynamic, the two roles being largely separate. Today, those roles are typically combined into a single one, that of the web developer, who must deliver high-quality and consistent UX through friendly and functional UIs thus demonstrating good design skills, while also knowing how to code the programming logic required to produce accessible and responsive data-driven web-based solutions meeting requirements.

You will learn to design and build modern web services and web applications using contemporary methods, tools, languages, frameworks, and standards. The module reflects current industry practice where web solutions are expected to be secure, responsive, accessible, and data-driven, with strong attention to performance and reliability. You will explore current UI/UX expectations alongside client-side development, server-side development, and database

access, while also engaging with wider issues surrounding web technology such as usability, accessibility, security considerations, and professional standards in web development.

This module is designed in a way that recognises that students may arrive with varied experience of web development and programming. Learning activities and practical tasks will provide structured support for core skills while offering opportunities for stretch and extension through increasingly complex, realistic development challenges. The emphasis is on building confidence through hands-on practice, frequent feedback, and iterative improvement, mirroring how web developers work in real teams, so that all students can progress from foundational concepts to producing robust, user-focused solutions. Also, opinions and ideas are openly and actively encouraged during these sessions.

The module builds upon skills and knowledge delivered in previous modules such as Databases and Introduction to Software Engineering to promote and build further understanding of the interlinked nature of various aspects of software development lifecycle and computing technology and how they work together to produce websites and web-based systems through the design, implementation and testing of a data-driven web-based solution using modern, industry-leading language(s) and design and development tools commonly used by industry. Additionally, it's one of the core modules which prepares the students for their project in the final year.

The module helps consolidate learning by requiring students to apply and combine previously learned concepts in an end-to-end, practical development context. Through designing, implementing, testing, and refining a complete web solution, students will strengthen their understanding of programming logic, data handling, and user-centred design while developing professional habits such as debugging, version control, documentation, and iterative enhancement.

4. Learning Outcomes and Assessments

Module Learning Outcomes	Assessment No. and Type
LO1: Assess and outline issues and concerns surrounding web technologies and make appropriate recommendations for their mitigation	Assessment 1
LO2: Analyse and evaluate an organisation's problems and requirements to design web-based solution with consideration of user experience (UX) and user interface (UI) design	Assessment 1
LO3: Develop, implement and test a data-driven web-based solution using appropriate and contemporary web-development techniques, languages, tools, and frameworks and reflect on own performance	Assessment 1

5. Assessment Deadlines

Assessment item		Due Date	Weight
1	Portfolio	23/04/2026 23:59	100%

6. Assessment Feedback

Feedback on items of assessment can be formal (such as on a signed feedback form) or informal (such as advice from a tutor in a tutorial). Feedback is therefore not just your grade or the comments written on your feedback form, it is advice you get from your tutor and sometimes your peers about how your work is progressing, how well you have done, what further actions you might take.

We recognise the value of prompt feedback on work submitted. Other than in exceptional circumstances (such as might be caused by staff illness), **you can expect your work to be marked and feedback provided not more than 5 working days** from the deadline date. However, please note that such feedback will be provisional and not agreed until the Assessment Board has met and may therefore be subject to change.

Please take time to read or listen to your assessment feedback. This can be very useful in determining your strengths and key areas for development, and can therefore help you improve on future grades.

7. Module Calendar

Session No.	Week Commencing	Topics Covered	Planned Delivery Method(s)
Week 1 Day 1	09/03/2026	Setup & Web Fundamentals: - Tooling setup (IDE and Git) - URL structure + Ports - Web protocols: HTTP/HTTPS, Request/response cycle - Browser Developer Tools introduction Assessment 1 briefing	On campus
Week 1 Day 2	09/03/2026	HTML, CSS & UI/UX Basics: - UI/UX principles - HTML/CSS (Tailwind) - Responsive design basics	On campus
Week 2 Day 1	16/03/2026	API Development with Express - REST & HATEOAS - Creating API routes - Request and Response Handling - Testing APIs with Postman - API documentation with Swagger/OpenAPI	On campus
Week 2		Application Architecture:	

Day 2		- Architecture patterns: Monolith vs N-tier - Structuring full-stack applications - Separation of concerns - Middleware	
Week 3 Day 1	16/03/2026	Backend Fundamentals: - PostgreSQL setup and basics - Introduction to Prisma ORM - Defining schemas in Prisma - CRUD operations: Create, Read, Update, Delete - Database migrations	On campus
Week 3 Day 2	23/03/2026	Connecting Backend to Database: - Integrating Prisma with Express routes - Implementing full CRUD operations with database - Error handling and validation - Testing database-connected APIs	On campus
Week 4 Day 1	23/03/2026	Frontend Development with React: - React fundamentals with TypeScript - Components, props, and state - React Hooks (useState, useEffect) - Building forms and handling user input - Connecting frontend to backend API (fetch/axios) - Updating frontend with API data	On campus
Day 2 Week 4	14/04/2026	Security Fundamentals: - Secure connections (touching on HTTPS), gateways/load balancers (Kong) - Authentication concepts (JWT, sessions) - Password encryption (bcrypt) - OWASP Top 10 security risks - Input validation and sanitization - Introduction to OWASP Juice Shop	On campus
Day 1 Week 5	21/04/2026	Testing Strategies - Testing pyramid overview - Unit testing with Jest - Component testing with React Testing Library - Integration testing basics - End-to-end testing with Playwright (light coverage) - Test-Driven Development (TDD) introduction	On campus
Day 2 Week 5	21/04/2026	Deployment & Production:	On campus

		- Deployment basics: hosting platforms overview (Vercel) - Deploying a Next.js/Express app (hands-on exercise) - Environment variables in production - Database hosting essentials - Introduction to monitoring and analytics tools	
		Project Demo	

8. Continuous Assessment Strategy

This module will be assessed through a portfolio of activities, which will include presentation, design document and implemented solution. Along with outlining, assessing, and suggesting approaches to mitigate issues and concerns surrounding web technologies, you will be given a scenario to analyse, design, develop, implement, test and review a web-based solution. You will produce this using appropriate software development techniques, languages, tools, and frameworks. You will also select and utilise an appropriate testing methodology to verify the requirements have been met and critically review and appraise the final outcome. The implemented solution will consist of a report and source code submission.

9. Indicative Reading – Essential and Recommended

Essential Reading:

Choi, D. (2025) *Full-stack React, Typescript, and Node: build scalable and cloud-ready web applications using React 19, Typescript, and Docker*. Birmingham: Packt Publishing.

Wieruch, R. (2025) *The road to React: the React.js 19 with Hooks in JavaScript book*. 2025 ed. Self-published.

Lauret, A. (2025) *The design of web APIs*. Second. Shelter Island: Manning Publications.

Hoffman, A. (2024) *Web application security: exploitation and countermeasures for modern web applications*. 2nd edn. Sebastopol, CA: O'Reilly Media.

Recommended Reading:

Vanderkam, D. (2024) *Effective TypeScript: 83 specific ways to improve your TypeScript*. 2nd edn. Sebastopol, CA: O'Reilly Media.

Costa, L.F.d. (2021) *Testing JavaScript Applications*. 1st edn. New York: Manning Publications Co. LLC.

Wathan, A. and Schoger, S. (2018) *Refactoring UI*. [Digital book]. Self-published.

Mammino, L. and Casciaro, M. (2024) *Node.js design patterns: design and implement production-grade Node.js applications using proven patterns and techniques*. 4th edn. Birmingham: Packt Publishing.

Brown, E. (2019) *Web Development with Node and Express, 2nd Edition*. O'Reilly Media, Inc.

10. Guidelines for the Preparation and Submission of Written Assessments

1. Written assessments should be word-processed in Arial or Calibri Light font size 12. There should be double-spacing and each page should be numbered.
2. There should be a title page identifying the programme name, module title, assessment title, your student number, your marking tutor and the date of submission.
3. You should include a word-count (excluding reference list/bibliography, figures, tables and appendices).
Where a word limit is specified, the following penalty systems applies (For HE5, HE6 & HE7 Modules only):
 - Up to 10% over the specified word length = no penalty
 - 10 – 20% over the specified indicative word length = 5 marks subtracted (but if the assessment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assessment).
 - More than 20% over the indicative word length = if the assessment would normally gain a pass mark or more, then the final mark will be capped at the pass mark for the assessment.
4. **At the end of the assessment you should include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment. The assessment brief will specify if and how you can use GAI applications in the assessment.**
5. All written work should be referenced using the standard University of Greater Manchester referencing style– see: <https://libguides.bolton.ac.uk/resources/referencing/>
6. Unless otherwise notified by your Module Tutor, electronic copies of assignments should be saved as word documents and uploaded into Turnitin via the Moodle class area. If you experience problems in uploading your work, then you must send an electronic copy of your assessment to your Module Tutor via email BEFORE the due date/time.
7. Please note that when you submit your work to Moodle, it will automatically be checked for matches against other electronic information, as well as for hidden text characters and GAI generated text. You will be able to see similarity matches but not currently flags for hidden characters and AI-generated text. The outcomes of Turnitin reports may be used as evidence in an academic misconduct investigation (see Section 14).

11 Late work

Late work will be subject to the following penalties:

- Up to 5 calendar days late = 10 marks (If HE5 & HE6) subtracted but if the assignment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assignment.
- More than 5 calendar days late = This will be counted as non-submission and no marks will be recorded unless this has been agreed with the Programme Leader, Module tutor or Personal Tutor.

Late submission of assessments on refer and those which are graded Pass/Fail only, is not permitted unless an extension is approved. See below.

12 Extensions

In the case of exceptional and unforeseen circumstances, an extension of up to 5 days after the assessment deadline may be requested using the standard University Extension Request Form. For approval there would need to be an explanation and evidence of relevant circumstances.

Longer extensions for individual assessments, projects and artefacts may be granted, at the discretion of the Programme Leader.

Requests for extensions which take a submission date past the end of the must be made using the Special Circumstances procedure.

Some students with registered disabilities will be eligible for revised submission deadlines. Revised submission deadlines for disability adjustments do not require the completion extension request paperwork. However, students should request these in writing in advance.

Please note that the failure of data storage systems is not considered to be a valid reason for an extension. It is therefore important that you keep multiple copies of your work on different storage devices before submitting it.

13 Procedures for Other Assessments

There are no alternative formal assessments in this module.

14 Academic Misconduct

Academic misconduct may be defined as any attempt by a student to gain an unfair advantage in any assessment. This includes plagiarism, collusion, commissioning (contract cheating and unauthorised use of GAI) amongst other offences.

In order to avoid these types of academic misconduct, you should **ensure that all your work is your own and that sources and software applications are attributed**. You can also check originality through *Turnitin*.

Please note that penalties apply if academic misconduct is proven. See the following link for further details:

<https://greatermanchester.ac.uk/student-policy-zone/student-policies-2025-26/academic-misconduct-regulations-and-procedures-2025-26>

15 Assessments

See the assessments briefs that are issued separately and are not included in this document.

16 General Assessment Criteria for Written Assessments

GENERAL ASSESSMENT GUIDELINES – LEVEL HE5

	Relevance Learning outcomes must be met for an overall pass	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/Referencing	Written English	Presentation and Structure
Class I (Exceptional Quality) 85% - 100%	Work is directly relevant and expertly addresses the requirements of the brief. Learning outcomes are met.	Demonstrates an exceptional breadth and depth of knowledge and understanding of theory and practice beyond the threshold expectation for the level. Demonstrates sophisticated conceptual understanding of specialised areas.	Presents a perceptive synthesis and critical evaluation of findings' drawing on a broad range of relevant sources in order to draw clear, systematic, justified conclusions. Provides a critical insight and interprets complex matters and ideas comprehensively. Demonstrates creative flair and a high level of originality. Demonstrates exceptional problem-solving skills and initiative.	Provides insightful reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of contemporary and relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An exceptionally well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class I (Excellent Quality) 70% - 84%	Work is relevant and comprehensively addresses the requirements of the brief. Learning outcomes are met.	Demonstrates an excellent breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates in-depth understanding of key areas in different contexts.	Presents an excellent appraisal of findings through the critical analysis of information sources. Draws clear, systematic, justified and perceptive conclusions. Demonstrates creative flair, a high level of originality and initiative. Demonstrates a critical understanding of problem-solving approaches and applies excellent problem-solving skills.	Provides excellent reflection and self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An excellently well written answer with competent, spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class II/i (Very Good Quality) 60% - 69%	Work is relevant and addresses most of the requirements of the brief. Learning outcomes are met.	Demonstrates a thorough breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates very good understanding of key areas in different contexts.	Presents a perceptive appraisal of findings through the critical analysis of information. Draws clear, justified and thoughtful conclusions. Demonstrates a very good level of creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies strong problem-solving skills.	Provides very good reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	A range of appropriate reference sources selected and drawn upon. Sources cited accurately in the main both the body of text and in the Reference List/Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. A very well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured in the main. Where relevant, effective inclusion of, and reference to, figures, tables and images.

	Relevance Learning outcomes must be met for an overall pass	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/Referencing	Written English	Presentation and Structure
Class II/ii (Good Quality) 50% - 59%	Work addresses key requirements of the brief. Some irrelevant content. Learning outcomes are met.	Demonstrates a sound breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates sound understanding of key areas in different contexts.	Presents a logical appraisal of findings through the critical analysis of information. Draws clear and justified conclusions. Demonstrates creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies effective problem-solving skills.	Provides valid reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Relevant reference sources selected and drawn upon. Most sources accurately cited both the body of text and in the Reference List/Bibliography.	Writing style is mostly appropriate to the requirements of the assessment. Grammar, spelling and punctuation are generally competent and minor lapses do not pose difficulty for the reader. Paragraphs are structured and include some linking and signposting. Sentences are complete. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Logically structured in the most part. Inclusion of figures, tables and images but not all relevant or referred to.
Class III (Satisfactory Quality) 40% - 49%	Work addresses the requirements of the brief, although superficially in places. Some irrelevant content. Learning outcomes are met.	Demonstrates a sufficient breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates understanding of a range of key areas in different contexts.	Presents valid appraisal of findings in parts following a critical appraisal of information. Interprets a number of matters and ideas but with some descriptive passages evident which lack clear purpose. Demonstrates creativity and originality in places. Demonstrates a critical understanding of problem-solving approaches and applies sufficient problem-solving skills.	Provides some reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Some relevant reference sources selected and drawn upon. Some weaknesses in referencing technique.	Writing style is occasionally not appropriate for the assessment. Grammar, spelling and punctuation are generally competent, but may pose minor difficulties for the reader. Some paragraphs may lack structure, and there is limited linking and signposting. Some appropriate vocabulary is used.	The presentational style and layout are largely correct for the type of assignment. Adequately structured. Inclusion of some figures, tables and images but not all clear, relevant and/or referred to.
Borderline Fail 35% - 39%	Work addresses some of the requirements of the brief. Irrelevant and superficial content. One or more learning outcomes have not been met.	Demonstrates a lack of knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of a range of areas in different contexts.	Presents a descriptive overview of findings following some analysis of information. Some passages lack clear purpose. Demonstrates scant creativity or originality. Demonstrates a lack of critical understanding of problem-solving approaches and insufficient problem-solving skills.	Provides limited reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	Sources selected are limited and/or lack relevance. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear in places. Paragraphs are poorly structured.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.
Fail <34%	Work does not address the requirements of the brief. Irrelevant and superficial content. One or more learning outcomes have not been met.	Demonstrates inadequate knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of relevant areas.	Presents a descriptive overview of findings with little consideration of the quality of information drawn upon. Demonstrates little or no creativity. Demonstrates a lack of understanding of problem-solving approaches and limited problem-solving skills.	Provides inadequate reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	An absence of academic and relevant sources selected and drawn upon. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear. Unstructured paragraphs.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.